THE NOVA RESEARCH CURRICULUM

Chemistry

Grade 8 · Term 1

Student edition

Outcome: CH-8.1 the periodic table (first 18 elements); metals and non-metals.

Lesson notes

- **The periodic table (Weeks 1–4):** elements arranged by atomic number. Learn the first 18: H, He, Li, Be, B, C, N, O, F, Ne, Na, Mg, Al, Si, P, S, Cl, Ar.



The first 18 elements. Rows are periods; columns are groups.

- **Groups and periods (Weeks 5–8):** a group (column) shares the same number of outer electrons, so similar reactions; a period (row) shares the same number of shells. Group 1 (alkali metals) react strongly with water; Group 17 (halogens) are reactive non-metals; Group 18 (noble gases) are unreactive because their outer shells are full. Teacher video only for alkali metals in water.
- **Metals and non-metals (Weeks 9–12):** metals: shiny, good conductors, malleable, ductile, mostly solid; non-metals: dull, poor conductors (except graphite), brittle when solid. Metalloids such as silicon. Uses in daily life.

Workbook pages

Page	Task
1	Fill in the first 18 elements with symbols and atomic numbers.
2	Group and period puzzle.
3	Metals vs non-metals property table and conductivity test.

Question bank

#	Outcome	Question	Your answer
1	CH-8.1	Elements in the periodic table are arranged by ...	name / colour / atomic number / discovery date
2	CH-8.1	Elements in the same group have the same number of ...	shells / neutrons / outer electrons / protons
3	CH-8.1	Which is a noble gas?	oxygen / neon / chlorine / sodium
4	CH-8.1	Noble gases are unreactive because ...	they are heavy / their outer shell is full / they are metals / they are liquids
5	CH-8.1	The symbol Na stands for ...	nitrogen / neon / sodium / nickel
6	CH-8.1	Metals are good conductors of electricity. True or false.	True / False
7	CH-8.1	Which non-metal conducts electricity?	sulfur / oxygen / graphite (carbon) / phosphorus
8	CH-8.1	Give two properties of metals.	_____